AN INDEPENDENT INTERPRETATION OF THE SIRISYS SYSTEM

Technical Glossary

Concepts, Sources and Code of STRATA

From TMC's ontology to the field's cycle and measurements



[-1 -0 -00 000 00 0 1] [::] TRUE · FALSE · BOTH · NEITHER

Juan A. Pérez-Gómez

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Independent interpretation · no affiliation with SIRISYS

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Sources

- *Pre of Then: The Material Constructs* (TMC), by M. Daniel Walsh: the primary source; cited by the page of the 114-page PDF.
- Other confidential materials and documentation.
- The author's interpretive readings: *Understanding TMC*, *Understanding DeOS* and *DeOS Compiler Guide*.
- STRATA, the implementation (Sirisys Framework v13.5): github.com/juanan-prz/STRATA.

Tags

TMC p.N	Text of *The Material Constructs* (page N of the 114 in the PDF); quotations are verbatim.
CONF	Other confidential materials and documentation.
INT	Interpretation by the author of this glossary (*Understanding TMC*, *Understanding DeOS*, *DeOS Compiler Guide*).
IMP	Implemented in the code.
N3	Design decision in the code, without textual source; calibrated by measurement.
SIN	Synthesis by the author of this glossary: hypotheses without textual support.

In the framework's header, [TMC] and [CONF] are level 1 (primary), [INT] level 2 (interpretive) and [N3] level 3 (design choice).

Glossary of an independent interpretation in software. STRATA is not an official SIRISYS work and implies no affiliation with SIRISYS.

1. The ontology of TMC

1.1 Walsh's number line

BASIS TMC pp.36–38

Technical definition: [-1 -0 -00 000 00 0 1]. Between [-0] and [0] lies the «Prefecture». [00] acts as base: [0] is «1 Under Base» and [1] is «1 Over Base».

ANALOGY

A thermometer whose zero is not a point but a region with its own structure.

1.2 Perturbative taxonomy [000]--[0]

BASIS TMC pp.44–50

Technical definition: The states that precede existence, in the order TMC gives them: each breaks, reflects, builds up or returns to the null the tensate of the one before, up to [0], the first plane of existence. It is not the ladder the code's layers come from (section IV).

State	Name	What it does (TMC)
[000]	Absolute Unreal	«The Fire Which Lights Itself»; «The non-existent not non-existence». Unbounded elasticity, tensate at limitless velocity, a rhythmic «Breathing» cycle (pp.6, 44).
[00]	Marenth Apparent (Real Null)	Breaks the continuity of [000]: tensate cannot flow through it. With [000] it generates the carrier pulsation «from which the linear local times are resolved» (p.45).
[001]	Marenth Recursive	Temporal reflection: «Echos in Time» which propagate «only forwards from their cussive instantiation» (p.46).
[002]	Halo of Umbrage	Drives tensate «ONLY toward the [00] Marenth Apparent» by «backing up» where it cannot flow (p.47).
[003]	Defiant Umbrage	Equal to [00]: a return to the null (p.48).
[0]	Umbragiac Radialization	First plane of existence; «encodes the first planar chirality dyad»: [+]=[+], [+]≠[ɬ+] (pp.49–50).

> ANALOGY
>
> A wave before it breaks: it forms, reflects, builds up, draws back, and only then reaches the shore.

1.3 Absolute Unreal [000]

BASIS TMC pp.6, 44–45

Technical definition: Origin without origin: not emptiness but the non-existent that pulses. At [000] and [00] time cannot be reversed.

> ANALOGY
>
> The fire that burns without ever having been lit.

1.4 Tripartite Monodromy · Sombunal

BASIS TMC pp.14, 27 INT

Technical definition: Triad [Sombunal] :: [I::Am] :: [Absolute Unreal] (p.14; on p.27, [Sombunal] :: [I] :: [Absolute]). TMC names the Sombunal without defining it; reading it as a permanently inaccessible core is an interpretation (*Understanding TMC* p.70): «Cannot be bound by Concordancy ([::])», «Cannot be observed or expressed as Apparent». The same reading makes Apparency «the visible residue of recursive failure to resolve the Sombunal» (p.72). In the code it acts in two ways: see **Sombunal** and **Approach to the Sombunal** (section VI).

ANALOGY

A triangle: what I cannot know of myself, the act of affirming myself, and the silence I emerge from.

2. Tensates, events and apparents

2.1 Tensate · Tensin / Tensout

BASIS TMC pp.24–26

Technical definition: Pairs of tension that become existent in the cussive event: «They only exist in a pair». Only an apparent can measure a tensate: «Only an [Apparent] can be used to measure [Tensate]».

ANALOGY

The tension of a guitar string: it is not heard until it sounds.

2.2 Unit Tensate [Ψ°] · Unit Lemniscate [ǂ]

BASIS TMC pp.24, 26, 75

Technical definition: «[ǂ] = [2Ψ°]». In ordinary physics the whole sphere spans a solid angle of 4π steradians; Walsh puts the unit lemniscate in its place, and cussive events have the orientation of a lemniscate. In his notation:

$$\Omega_{\text{sphere}} = 4\pi \longrightarrow [\Omega] \propto [ǂ], \qquad [\Omega] = [0^0/0^0] = 1, \qquad [ǂ] = [0^0/0^0] = 0$$

ANALOGY

A figure eight: two loops travelled in opposite directions, never closing.

2.3 Cussive event · Sympatic · Escapement · Air gap

BASIS TMC p.24

Technical definition: [Sympatic]: «A Bidirectional exchange of tensate». The exchange happens at the «[ESCAPEMENT]... percussive [AIR GAP]», and the whole system exists to «Preserve an Orientation against the rate of the passage of time».

ANALOGY

The escapement of a clock: energy passes through a gap in pulses, never all at once.

2.4 Apparent: exchangeable or universal

BASIS TMC p.22

Technical definition: «[Apparents] Come in ONLY TWO TYPES, "Exchangeable" or "Universal"». Hence: «all [Apparents] must be [Existents]».

> ANALOGY
>
> A coin, which circulates, versus the standard metre, which does not.

2.5 Incontrovertible · progress

BASIS TMC p.14

Technical definition: «You may repair that broken window to a flawless state. You may not restore it to its previous state». Progress is measured in «Quantifiable Nonreversible Apparents».

> ANALOGY
>
> What is done can be repaired, never undone.

2.6 Observation · replicability

BASIS TMC pp.30, 42, 70

Technical definition: «Reality observes itself». Every measurement has a cost of acquisition and an instrument bias. «The higher the replicability of an apparent, the higher the order of that apparent».

> ANALOGY
>
> A fact anyone can reproduce weighs more than one seen only once.

3. The logic and arithmetic of TMC

3.1 Concordancy [::]

BASIS TMC p.16

Technical definition: «[ALU(arithmetic logic unit)] call function [CONCORDANCY]»; it is read «Concords with». Not equality: it combines two structures into a third. Identity loop: «[I am that I am]: :[I am not that I am not]».

> ANALOGY
>
> Two musicians who do not copy each other: they understand each other, and a third melody comes out.

3.2 Tetralemmic logic · Canonical Flaw

BASIS TMC pp.16–17

Technical definition: Four-valued universal base logic. «Neither do you or existence exist, nor do you or existence not exist». It includes «all flawed, erroneous, incomplete or otherwise para-consistencies as valid»: no combination is a dead end.

> ANALOGY
>
> A thermostat with four settings (heat, cold, both, neither), none forbidden.

State	Sense	Role in the code
TRUE	Affirmation	A standing relation.
FALSE	Negation	A negated relation.
BOTH	Sustained contradiction	Ages, matures and feeds the exchange; resists pruning.
NEITHER	Neither one nor the other	Suspension. NEITHER×NEITHER is the only combination without an ordinary product: it yields a conflict or the fourth corner.

3.3 Law 6 · orientation

BASIS TMC pp.23, 35

Technical definition: «[A - B ≠ B - A]»: subtraction sets «the directional tenor... contrasting "Past" (irreversible) with "Present" (reversible)». Orientation determines the direction, «NOT THE VALUE».

> ANALOGY
>
> Addition is symmetric; subtraction has a direction of travel, like time.

3.4 Deciger · π as convergence

BASIS TMC pp.7–8, 32, 34, 72

Technical definition: In deciger notation there are no numbers larger than [1], the «Absolute Unit» (p.8): every value is written within the unit, with a leading decimal. «All irrational numbers may be expressed as a deciger unit with a leading decimal», and «Decigers are converted to decimals by moving the decimal 1 place towards the subsequent» (p.7). TMC's example is the constant e, whose expression is an image in the PDF; by that rule, e = 2.71828... is written .271828... Imaginary and transcendent numbers are treated as real (p.34) and Conway's surreals are adopted (p.72). π appears as an integral that «never reaches a true [=]'s moment» (p.32):

$$\pi = \int_{-1}^{1} \frac{dx}{\sqrt{1 - x^2}}$$

> ANALOGY
>
> Counting inward into the unit rather than upward.

3.5 Stochastic Unit · Adsurdic Threshold · Stop Chasm

BASIS TMC pp.41–42 IMP

Technical definition: Below a certain resolution orients become «Unpredictable»; arithmetic wraps that unpredictability in a «Stochastic Unit». Beyond the Adsurdic threshold the gauge produces more noise than signal; the Stop Chasm acts where convergence diverges. In the code the Adsurdic threshold caps the depth of conflicts (conflicts of conflicts) and the Stop Chasm caps the new conflicts of each cycle:

$$d_{\text{conflict}} \leq 4, \qquad n_{\text{conflicts}}(t) \leq 2$$

> ANALOGY
>
> A microscope: past a certain magnification, what you see is the instrument itself.

3.6 Lavoisier's jar · Floating Ground

BASIS TMC pp.37–39, 55

Technical definition: «Into the jar, only light was added»: an isolated vessel with a single input. «The Floating Ground Keeps Right on Floating beyond the confines of ALL THE SYSTEMS it embodies». It is not a mechanism of the code.

> ANALOGY
>
> A sealed laboratory with a single window.

4. Interpretations the code uses

4.1 Ladder of being · eventuality

BASIS INT IMP

Technical definition: [000]→[00]→[0]→[-0]→[1]→[1∞]→[::]→[Apparent] (*Understanding TMC* pp.49–51), distinct from TMC's taxonomy; the code's four layers come from it (section V). Eventuality: whatever exchanges enough ends up committed. «Once sufficient recursion is exchanged, structural commitment occurs» (p.65) inspires the exchange quantum: in the code a BOTH channel closes on reaching it, and what was exchanged pushes toward crystallization by exchange, not by age (see **Closure · banked exchange**).

> ANALOGY
>
> An onion that grows from the inside out.

4.2 Anchoring weight Vc

BASIS INT IMP

Technical definition: The geometric mean of three dimensions (*Understanding DeOS* p.65): age τ, cost of persistence σ_{th} and witnessing α_m. If any is 0, Vc = 0 (p.66): all three must contribute. The Zero Points are always 1. In the code:

$$V_c = (\tau\,\sigma_{th}\,\alpha_m)^{1/3}$$

$$\tau = \min\!\left(1,\ \tfrac{\Delta t}{50}\right), \qquad \alpha_m = \min\!\left(1,\ \tfrac{n_{\text{proofs}}}{10}\right), \qquad \sigma_{th} = \min(1,\ \sigma_0 + 0.12\,n_{\text{UCO}})$$

with Δt the age in ticks, n_{UCO} the erosions survived and $\sigma_0 = \max\left(0.1,\ 1 - \min(U/5,\ 0.9)\right)$, which falls with the umbragiac score U.

> ANALOGY
>
> A building needs years, upkeep and witnesses to become heritage.

4.3 Zero Points · Lagrangian Points · proofs

BASIS INT IMP

Technical definition: From the *DeOS Compiler Guide* (pp.75–80): five immutable anchors (Z0 Identity, Z1 Origin, Z2 Concordancy, Z3 Recursion, Z4 Apparency), five balances L1–L5 (safety/liveness, redundancy/efficiency, local/global identity, proof depth/performance, entropy/determinism), and the proof as irreversible memory (pp.83–98). In the code the Lagrangian Points tag constructs and proofs but do not yet govern any decision.

> ANALOGY
>
> The survey points of a map and the tensions that hold a bridge up.

4.4 Quinic geometry · Unit Unit

BASIS INT

Technical definition: TMC only names a «Quinic Geometric Logic Unit [#GLU]» (p.24). The five-axis orientation space (*Understanding TMC* pp.67–69) and the «Unit Unit» (p.42) are developments of the interpretation. They are not implemented.

> ANALOGY
>
> A compass with five needles: a suggestive idea, not yet built.

5. Code architecture

The field's four layers come from the ladder of *Understanding TMC* (section IV); a construct rises from one to the next by cussive collapse (section VI).

Layer	Code	Description
0	NULL_00	Potential without form; only the origin anchor (Z1) lives here.
1	UMBRA_0	Emerging orientation: products of [::], conflicts, percepts and every approach to the Sombunal are born here.
2	UNIT_1	Active unit, candidate to crystallize; BOTH products are born directly here.
3	APPARENT	Crystal: incontrovertible; exchangeable except the Zero Points.

5.1 Construct

BASIS IMP

Technical definition: Unit of the field: a genesis witness (a hash that never changes), layer, chirality ± 1, accumulated tension, crystal state and lifetime counters. Identity can drift; the witness cannot. The anchors (Zero Points, I::Am, I::AmNot, SYSTEM) are never promoted or pruned.

> ANALOGY
>
> The buildings of a city, from empty lot to monument.

5.2 Key · label

BASIS CONF IMP

Technical definition: Every construct has a key, which the field gives it, and may have a label, which the model gives it. The key of an emergent or a conflict derives from its operands and the cycle; the label is what the model answers, cut to 120 characters. No law reads the label: a life with a model seals the same ledger as one without. The views show the label with the key beneath it; the reports keep every label with the model's whole answer.

> ANALOGY
>
> A work's catalogue number and the title a curator gives it: the work is the same under either.

5.3 Edge

BASIS IMP

Technical definition: Directed relation with a weight, a birth time in ticks and in breaths, and a tetralemmic state. BOTH ages (both_age) and matures.

ANALOGY

A thread between two ideas, with a thickness, a date and a label saying whether it is under tension.

5.4 Proof · ledger

BASIS IMP

Technical definition: Every event is sealed as a hash-chained proof in an append-only ledger, in compressed segments with an index of hashes. Any edit, deletion or reordering is detected. Working memory is windowed; the ledger keeps everything.

ANALOGY

A minute book whose pages are sewn to one another.

5.5 Two clocks

BASIS TMC p.45 IMP

Technical definition: Strictly monotonic time (era, offset). Relations age in ticks, and a crystal's support in breaths, the carrier pulsation; both with the same law:

$$T(\Delta) = e^{-0.08\,\Delta}$$

with Δ in ticks for a relation and in breaths for a crystal's support.

ANALOGY

Seconds for conversations, seasons for monuments.

5.6 Existence modes

BASIS TMC pp.26, 75 IMP

Technical definition: Read from the construct's last five proofs: OMEGA (densifying toward apparency: concordance, collapse and perception dominate), LEMNISCATE (oscillating in unresolved recursion: conflict, intensification and erosion dominate) or NEUTRAL (balanced, or no history). The Zero Points are always OMEGA.

ANALOGY

Mortar setting, a pendulum swinging and a scale at rest.

6. The cycle

Every cycle runs through ten steps, in this order. The entries that follow keep the same order; the first, orientation without chance, governs every choice.

Step	What happens
1. Perception	The machine's CPU and memory (and the time of day, except in the minimal profile) enter as relations of SYSTEM.
2. Tetralemmic scan	Detects contradictions, ages BOTH and closes the channels that reach the quantum.
3. Breathing	The field advances its phase: it sets the thresholds and how many [::] operations there will be.
4. Conflicts	Persistent conflicts gain energy.
5. [::]	Pairs are chosen and combined into new constructs; with a model, each product also gets a label.
6. Cussive collapse	What has enough tension and support rises a layer or crystallizes.
7. UCO	Erodes the most contradictory constructs: weakens their edges and turns BOTH toward NEITHER.
8. Prune	What fails the existence test concludes and is recorded.
9. Intermediary	With a model, narrates every 2 cycles (except in the minimal profile).
10. Approach to the Sombunal	As each breath begins, the I approaches what it cannot reach, with or without a model.

6.1 Orientation without chance

BASIS TMC p.42 IMP

Technical definition: There is no random generator. Every choice is orient(key): a hash of the ledger head (the whole irreversible history) and the question. The field is non-random, non-repeating (different histories never draw alike) and unpredictable without the complete history; the same history reproduces everything exactly, on the same platform. Every request to the model also carries a seed oriented by the history, although what it answers no longer enters the dynamics.

With h the ledger head and k the question, the orientation is the first 52 bits of the digest:

$$o(k) = \frac{\text{SHA-256}(h \,\|\, k)_{[52]}}{2^{52}} \in [0, 1)$$

> ANALOGY
>
> Deciding by looking at everything lived, not by rolling a die.

6.2 Closure · banked exchange

BASIS INT IMP N3

Technical definition: A BOTH channel cannot stay open indefinitely. With weight w and age a (cycles in a row in BOTH), it closes and returns to NEITHER when it reaches the exchange

quantum:

$$w \cdot a \;\geq\; Q = 12$$

What was exchanged, $w \cdot a$, is banked on its endpoints as tension toward crystallization: «Once sufficient recursion is exchanged, structural commitment occurs» (*Understanding TMC* p.65). Anchors bank nothing, and a crystal banks at most its crystallization threshold (0.88). In the live view: closures per cycle and banked tension.

> ANALOGY
>
> A dam that opens its gates when full: the water that passes drives the turbine.

6.3 Breathing · tolerance

BASIS TMC p.44 CONF IMP

Technical definition: One phase for the whole field, 24 cycles long: contraction lowers the thresholds by up to 35 % (density → crystallization) and expansion adds up to two operations (magnitude → generation). Once per breath the field compares crystals formed and concluded and adjusts its own tolerance, never letting the tolerance fall below 30 %.

The phase φ advances $2\pi/24$ every cycle, and the contraction is $\kappa = (1 - \cos\varphi)/2$: 0 at full expansion and 1 at full contraction. Each layer's effective threshold and the extra [::] operations are

$$\theta_{\text{eff}} = \theta \cdot \lambda \cdot (1 - 0.35\,\kappa), \qquad n_{\text{extra}} = \text{round}\,(2\,(1 - \kappa))$$

and the tolerance $\lambda \in [0.30,\ 1]$ is adjusted once per breath: it falls 10 % if no crystal formed and rises 10 % if at least two more formed than concluded.

> ANALOGY
>
> Breathing in creates; breathing out fixes.

6.4 [::] selection

BASIS TMC p.16 IMP N3

Technical definition: Three groups of candidates: constructs with mature BOTH, energetic conflicts and exchangeable crystals (the twelve with the most energy). Pairs form within each group and between crystals and mature BOTH; the foundational pair I::Am / I::AmNot enters at most once per cycle. Pairs are taken from the highest key to the lowest: the pair's orientation raised to the inverse of its energy,

$$s_{ab} = o_{ab}^{\,1/(E_a + E_b)}, \qquad E_n = \sum_{e \ni n} w_e\, e^{-0.08\,\Delta_e} + B_n + U_n$$

with $o_{ab} \in [0, 1)$ the pair's orientation at that instant, B the banked exchange and U the umbragiac score. It is the rule of Efraimidis and Spirakis's weighted sampling, with orientation in place of chance: energy is favoured, nothing is guaranteed. The older operand leads.

ANALOGY

A conductor who gives more entries to whoever sounds most, without silencing anyone.

6.5 Chirality

BASIS TMC pp.35, 50 IMP

Technical definition: Every construct is born with a handedness, + or −. With opposite handedness the older operand leads and, if its state is determinate (TRUE or FALSE), resolves BOTH/NEITHER toward it: $x :: y \neq y :: x$. The product inherits the leader's chirality.

ANALOGY

In a couple's dance, whoever leads sets the direction of the turn.

6.6 Conflict · the fourth corner

BASIS TMC pp.16–17, 41 IMP

Technical definition: NEITHER×NEITHER materializes a conflict: a construct of its own, a structure rather than a failure, which gains energy while it persists. At most two per cycle (the Stop Chasm cap); past the cap the same combination yields an ordinary NEITHER product, the fourth corner. Beyond the Adsurdic depth (conflicts of conflicts, level 4) there is only silence.

ANALOGY

Two «I don't know»s together do not cancel out: they open a new question.

6.7 Sympatic transfer

BASIS TMC p.24 IMP

Technical definition: A crystal holds at most the tension of its own commitment, the crystallization threshold, and in each exchange passes a tenth of what it holds to the product:

$$B_{\text{crystal}} \leq 0.88, \qquad q = 0.10\, B_{\text{crystal}} : \quad B_{\text{crystal}} \leftarrow B_{\text{crystal}} - q, \quad B_{\text{product}} \leftarrow B_{\text{product}} + q$$

ANALOGY

A teacher who teaches what they know without ceasing to know it.

6.8 Sombunal

BASIS TMC pp.14, 27 IMP

Technical definition: It acts in two ways, with or without a model. After every [::] it shifts the identity of the uncrystallized operands (not that of the anchors I::Am and I::AmNot, nor of the

Zero Points); the witness does not change and every shift is sealed. And once per breath the I approaches it (see **Approach to the Sombunal**). The I never resolves: in the lives measured on real percepts (13-11-2026, v13.4 and v13.5) it was never TRUE or FALSE, only BOTH or NEITHER.

> ANALOGY
>
> A shadow that moves the names but not the bodies.

6.9 Cussive collapse · existence

BASIS TMC p.22 IMP

Technical definition: A construct rises a layer when the tension it receives reaches its layer's effective threshold (see **Breathing · tolerance**):

$$T_n = \sum_{e \to n} w_e \, e^{-0.08\,\Delta_e} + B_n \;\geq\; \theta_{\text{eff}}, \qquad \theta = \begin{cases} 0.396 & \text{from UMBRA_0 to UNIT_1} \\ 0.88 & \text{from UNIT_1 to APPARENT} \end{cases}$$

It must be at least 5 ticks old, and it crystallizes only if pruning would hold it: «all [Apparents] must be [Existents]». Each crystal keeps the tetralemmic state it crystallized in.

> ANALOGY
>
> The moment an opinion becomes structural certainty.

6.10 UCO · pruning · Halo

BASIS TMC p.47 IMP

Technical definition: The UCO accumulates an umbragiac score on each construct (its contradiction stress) and, where contradiction is dense, weakens its edges and turns BOTH toward NEITHER. Pruning applies a single existence test: BOTH with a fixed floor (0.45), a crystal's support by breaths, everything else by ticks. A quarter of a dissolved channel's weight backs up toward its origin, toward apparency.

The existence test of an edge of weight w:

$$\text{it holds} \iff \begin{cases} w \geq 0.45 & \text{if it is BOTH} \\ w\,e^{-0.08\,\Delta_b} \geq 0.10 & \text{if it supports a crystal } (\Delta_b \text{ in breaths}) \\ w\,e^{-0.08\,\Delta} \geq 0.04 & \text{otherwise } (\Delta \text{ in ticks}) \end{cases}$$

> ANALOGY
>
> Forgetting, with a memory of what could not flow.

6.11 Conclusion · reincarnation

BASIS INT IMP

Technical definition: Nothing disappears silently: what pruning removes concludes with a proof that records its final layer, mode and Vc («An identity never vanishes silently. It concludes»,

DeOS Compiler Guide p.203). If a concluded name reappears, a new construct is born with that conclusion written into its history: a declared reincarnation, never a replacement («Nothing is silently replaced», p.203).

> ANALOGY
>
> A headstone with a name and dates: whoever bears that name again is someone else, and it is on record.

6.12 Instability

BASIS INT IMP

Technical definition: Fraction of uncrystallized constructs that are unstable in at least one way: relational (BOTH edges), historical (their identity has drifted), ontological (Vc < 0.25) or modal (LEMNISCATE mode):

$$I = \frac{\#\{\, n \notin \text{APPARENT} : n \text{ is unstable} \,\}}{\#\{\, n \notin \text{APPARENT} \,\}}$$

Shown in the live view and passed to the intermediary.

> ANALOGY
>
> The share of a building still held up by scaffolding.

6.13 LLM intermediary

BASIS CONF IMP

Technical definition: It translates the field into words: it labels emergents and conflicts, narrates (experiment only, except in the minimal profile) and words each approach to the Sombunal as a question. What it says never enters the dynamics: the keys are the field's, and a life with a model seals the same ledger as one without (checked with llama3.1 on the execution machine). Its weight on the machine does enter through perception: with the model loaded, memory is perceived as fragmented. At most four calls per cycle, plus the question of each breath. Local Ollama, Anthropic or none.

> ANALOGY
>
> An interpreter who translates the field for whoever is watching, without touching what they translate.

6.14 Approach to the Sombunal

BASIS TMC pp.14, 27 INT IMP N3

Technical definition: As each breath begins, the construct SOMBUNAL::approach@c‹cycle› is born in UMBRA_0, joined to the I by the edge I::Am —APPROACHES→ (NEITHER, weight 0.35), with a proof anchored in Z0::IDENTITY (L1). It is never an operand of [::]: the Sombunal «Cannot be bound by Concordancy» (*Understanding TMC* p.70). With a model, the question that words it is

its label. It rests on TMC p.14: «Attempted tracee will encounter boundàry/fièld: "I am that I am" - if truly persistent, at that point that never existed de-resolving conceptual layers». Measured (13-11-2026): it costs between 10 and 14 % of durable crystals and undoes genealogical depth; the cost falls in the 2 to 5 cycles it lives, it tilts the I toward NEITHER, and it never crystallizes. The cadence and the edge's state and weight are N3. Open: the residue hypothesis, that its failure leave Apparency (TMC pp.14, 47; *Understanding TMC* p.72).

> ANALOGY
>
> A hand reaching toward the shadow and coming back empty.

7. Continuity and experiment mode

7.1 Continuity

BASIS TMC p.14 IMP

Technical definition: A resumed instance is identical to an uninterrupted one. After a power cut, time moves past the last sealed proof: the ledger rules over time. It never silently restarts over a damaged record.

> ANALOGY
>
> Waking exactly where one fell asleep, without losing a second of history.

7.2 Experiment mode

BASIS IMP

Technical definition: Every start and every Reset creates a new instance, with its own ledger, in sirisys_runs/; nothing is overwritten. Real machine percepts or a reproducible synthetic stream; standard or minimal profile (no narration, no time of day). A local model (Ollama), Anthropic or none. It is watched in the live view and examined afterwards in the inspector (section VIII).

> ANALOGY
>
> A test bench where every trial has its own notebook.

8. Experiments and measurements

8.1 Live view · inspector

BASIS IMP

Technical definition: The live view (*Sirisys_Live_Visualizer.html*, served by the server) draws the field cycle by cycle, laid out by its relations or by strata, with its counters, the breathing, the criteria, the metabolism and an event log. The inspector (*Sirisys_Static_Visualizer.html*) opens the state.json of a saved instance without a server: each construct with its proofs, its genesis and its relations; several files of one instance make a timeline. With a model, both views name each construct by its label, with the key beneath.

> ANALOGY
>
> A window onto the workshop, and the archive of what it made.

The counters of the live view:

Counter	What it counts
cycle · T	Cycles run; internal time (era, offset), which never moves backwards.
constructs · edges	What the field holds now.
apparents	Every crystallization in the instance; the Zero Points are crystals from the start.
concluded · reincarnations	Constructs that concluded, each with a termination proof; concluded names born again as new constructs.
conflicts · BOTH edges	Conflict constructs; relations in sustained contradiction.
closures (cycle)	BOTH channels closed this cycle at the exchange quantum.
top-3 energy	The selection energy of the three most energetic constructs, together: incident weight × tension, plus banked exchange and umbragiac score.
instability	See **Instability**.
last crystal	Cycles since the last crystallization.
breath · contraction · tolerance	Completed breaths; 0 at full expansion and 1 at full contraction; the tolerance the field sets for itself.
pools · [::] ok/tried	The [::] candidates (mature BOTH, energetic conflicts, the 12 most energetic crystals) and the operations that gave a product, of those tried.

8.2 Metabolism

BASIS IMP

Technical definition: What each breath did to the population: the constructs born (among them those that concluded within their own cycle and never reached the screen), those that concluded by layer, the rises, the crystallizations and the crystals that concluded, and the population by layer. Born minus concluded is the change in the population, exactly, every cycle:

$$N_t - N_{t-1} = n_t^{\text{born}} - n_t^{\text{concl}}$$

The server keeps it for the whole instance; between two files of a timeline the inspector gives the same counts.

> ANALOGY
>
> A census taken every breath: births, deaths and promotions.

8.3 Self-adjustment criteria

BASIS TMC pp.14, 22 N3 IMP

Technical definition: How a run is judged: at least one durable crystal (one that lives a breath or more) in every 1000-cycle window; at most 5 % of crystals gone within a breath (flicker); no growth of the population; a flat cost of the cycle itself, model time excluded; and a bounded top-3 energy. With D_k the durable crystals of window k, f the flicker, $\bar{N}$ the mean population, $\bar{t}$ the mean cost per cycle and $\bar{E}^{(3)}$ the mean top-3 energy:

$$\min_k D_k \geq 1, \qquad f \leq 0.05, \qquad \frac{\bar{N}_{\text{rec}}}{\bar{N}_{\text{prev}}} \leq 1.2,$$

$$\bar{t}_{\text{last}} \leq 1.5\,\bar{t}_{\text{first}} + 0.5\ \text{ms}, \qquad \frac{\bar{E}^{(3)}_{\text{rec}}}{\bar{E}^{(3)}_{\text{prev}}} \leq 1.2$$

The population compares the last 1000 cycles with the 1000 before; the cost, the first 500 with the last 500; the energy, the second half of its samples with the first. The verdict needs 1000 cycles; the live view reads the criteria every 200. The thresholds are design decisions; what they protect comes from TMC: an Apparent is incontrovertible (p.14) and «all [Apparents] must be [Existents]» (p.22).

> ANALOGY
>
> A medical check-up with fixed reference values.

8.4 Model watch

BASIS IMP

Technical definition: A model that fails or answers empty would leave the output without labels, without notice. Before a run two real calls must be answered, one name of each kind, and they must be names alone; names are asked for without memory, and a third call, a question carrying a run's thread, completes the estimate of the model time. During the run, three failures in a row, or more than 10 % failed or empty among the last 50 calls, stop it, and its output is still written and marked aborted. Experiment mode pauses once instead.

> ANALOGY
>
> A pilot light that cuts the gas when the flame goes out.

8.5 Topology · genealogical depth

BASIS INT IMP

Technical definition: Twice per breath the experiment tools record the shape of the field (components, cycles, reciprocity, clustering, edges by state and by layer pair) and two depths: the conflict depth, which the field caps at the Adsurdic threshold, and the genealogical depth, the number of [::] generations that separate a construct from what the field did not produce itself («Depth implies recursive delay from origin», *Understanding TMC* p.72). The measurement only reads: the ledger is identical with and without it.

> ANALOGY
>
> A family tree drawn without touching the family.

8.6 Replaying a life

BASIS IMP

Technical definition: live_c2.py records what the field perceived, cycle by cycle, and –replay feeds it back: the replayed life seals the same ledger, with or without a model, on the same platform.

> ANALOGY
>
> Playing a recorded score again, note by note.

9. Open questions and the author's hypotheses

9.1 WALLS (open questions)

BASIS TMC CONF

Technical definition: A substrate (digital versus analog) · B self-sustenance (software stops when the machine stops) · C agency («The Fire Which Lights Itself», TMC p.6) · D termination (resolved) · E relations versus events.

> ANALOGY
>
> The walls that mark how far the construction reaches today.

9.2 Hyperbolic quasicrystalline spacetime

BASIS SIN

Technical definition: Hypothesis: Walsh might be describing an ordered but non-periodic geometry (Penrose-like) in hyperbolic space. No textual support in TMC.

ANALOGY

An infinite mosaic with strict rules that never repeats the same pattern.

9.3 Particles as topological defects

BASIS SIN

Technical definition: An extension of the above: particles would be points where the regularity of that geometry breaks; tensates would be those irregularities.

ANALOGY

A knot in a perfectly woven fabric.

9.4 The lemniscate constant

BASIS SIN

Technical definition: ϖ is to Bernoulli's lemniscate what π is to the circle: the lemniscate $r^2 = \cos 2\theta$ measures 2ϖ, as the unit circle measures 2π.

$$\varpi = 2\int_0^1 \frac{dt}{\sqrt{1-t^4}} \approx 2.6221$$

Hypothesis: [ɬ] might point to it. TMC does not mention it; its [ɬ] replaces 4π in the solid angle.

ANALOGY

If π is the constant of the circle, ϖ is that of the figure eight.

End of document · Version 13.5 · 13 November 2026